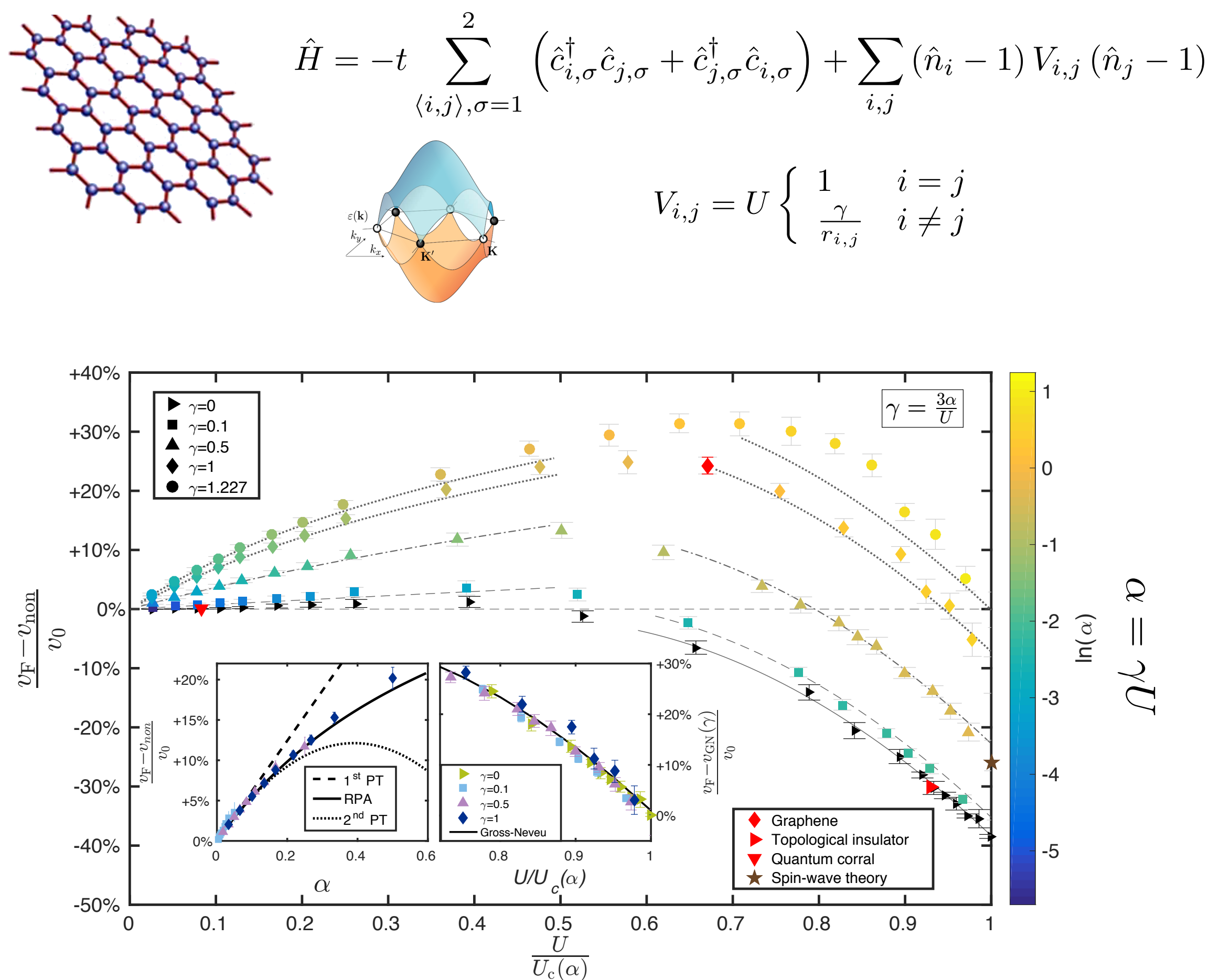


Goals:

- ✓ Generic package for auxiliary field Monte Carlo simulations of fermion-boson lattice models
- ✓ Efficient implementation on modern HPC systems
- ✓ Benchmarking and golden standards

The role of electron-electron interactions in two dimensional Dirac fermions



Ho-Kin Tang, J. N. Leaw, J. N. B. Rodrigues, I. F. Herbut, P. Sengupta, F. F. Assaad, and S. Adam, Science 361 (2018), no. 6402, 570–574.

PHYSICAL REVIEW LETTERS **120**, 107201 (2018)

Quantum Monte Carlo Simulation of Frustrated Kondo Lattice Models

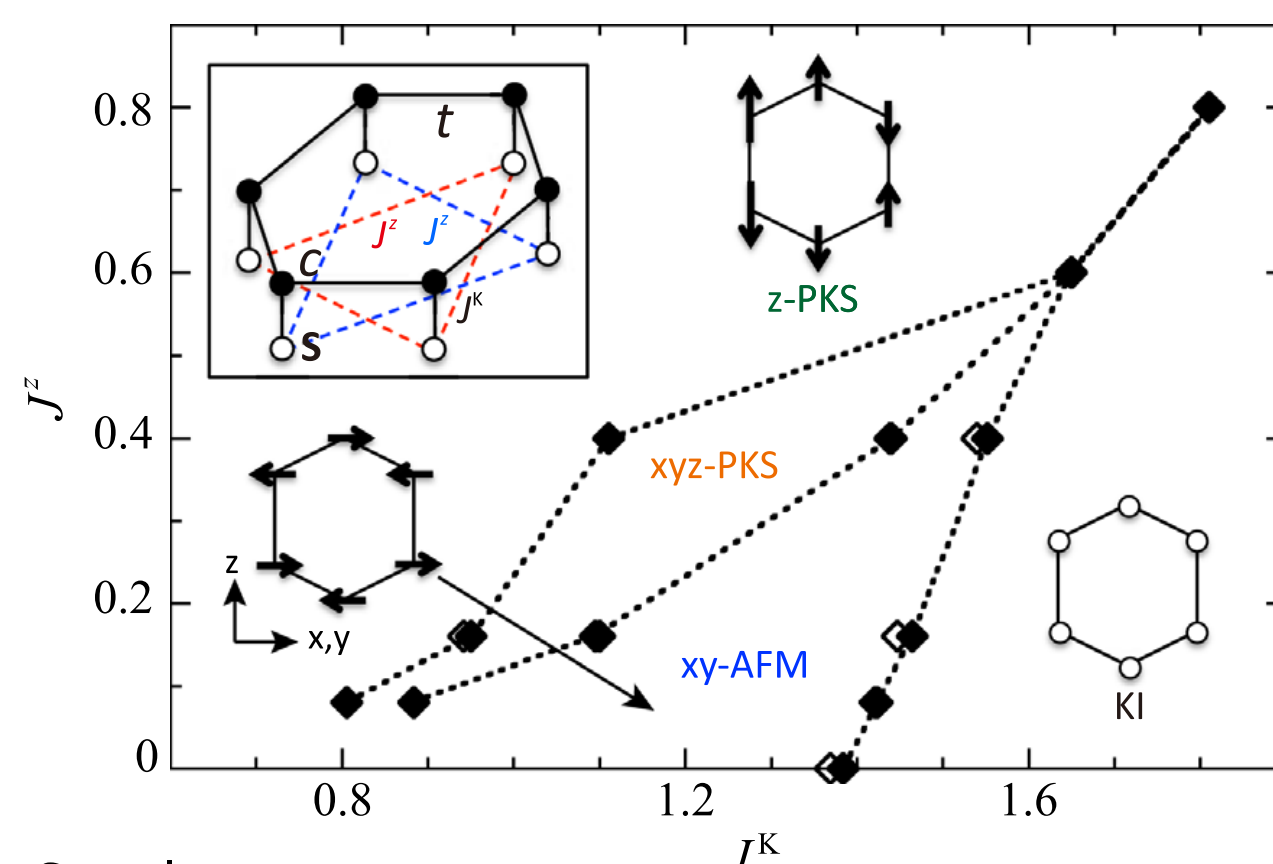
Toshihiro Sato,¹ Fakhre F. Assaad,¹ and Tarun Grover²

$$\hat{H} = \hat{H}_{\text{Spin}} + \hat{H}_{\text{Fermion}} + \hat{H}_{\text{Kondo}}$$

$$\hat{H}_{\text{Spin}} = \sum_{i,j} [J_{ij}^z \hat{S}_i^z \hat{S}_j^z + J_{ij}^\perp (\hat{S}_i^+ \hat{S}_j^- + \text{H.c.})],$$

$$\hat{H}_{\text{Fermion}} = \sum_{x,y,\sigma} \hat{c}_{x\sigma}^\dagger T_{x,y} \hat{c}_{y\sigma} + \sum_x U \left(\hat{n}_{x,\downarrow} - \frac{1}{2} \right) \left(\hat{n}_{x,\uparrow} - \frac{1}{2} \right),$$

$$\hat{H}_{\text{Kondo}} = \sum_{i,x} \frac{J_{i,x}^K}{2} \hat{c}_x^\dagger [\sigma^z \cdot \hat{S}_i^z - (-1)^x (\sigma^+ \hat{S}_i^- + \sigma^- \hat{S}_i^+)] \hat{c}_x.$$



Interplay of Kondo, RKKY
and geometric frustration

See also.

J. S. Hofmann, FFA, and T. Grover, PRB 2019, FL* phase in Kondo lattice model. BFG model Kondo coupled to electrons.

M. Raczkowski and FFA, PRL 2019, B. Danu, FFA, and F. Mila, arXiv:1903.08622 Nano-Kondo systems.

The ALF package

Kinetic

$$\hat{H} = \sum_{k=1}^{M_T} \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \sum_{x,y} \hat{c}_{x\sigma s}^\dagger T_{xy}^{(ks)} \hat{c}_{y\sigma s} + \sum_{k=1}^{M_V} U_k \left\{ \sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \left[\left(\sum_{x,y} \hat{c}_{x\sigma s}^\dagger V_{xy}^{(ks)} \hat{c}_{y\sigma s} \right) + \alpha_{ks} \right] \right\}^2 + \sum_{k=1}^{M_I} \hat{Z}_k \left(\sum_{\sigma=1}^{N_{\text{col}}} \sum_{s=1}^{N_{\text{fl}}} \sum_{x,y} \hat{c}_{x\sigma s}^\dagger I_{xy}^{(ks)} \hat{c}_{y\sigma s} \right) + \hat{H}_{\text{Ising}}$$

Potential (sum of perfect squares)

Coupling of fermions to Ising field

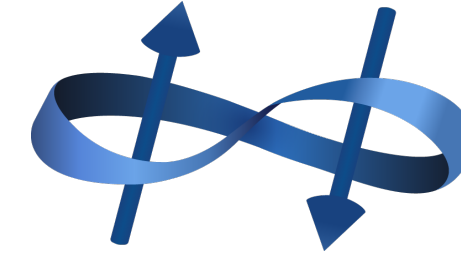
- Model can be specified at minimal programming cost
- MPI/OpenMP implementation
- Fortran 2003 standard
- SU(N_{col}) symmetric in colors N_{col}
- Arbitrary Bravais lattice for d=1, 2
- Block diagonal in flavors, N_{fl}
- Parallel tempering, projective and finite T approaches

Future/ongoing developments

- Continuous fields
- Hybrid molecular dynamics
- Langevin dynamics
- Imaginary time dependent Hamiltonians



Wissenschaftliche
Literaturversorgung
und Informationssysteme (LIS)



SFB1170
ToCoTronics

M. Bercx, F. Goth, J. S. Hofmann and F. F. Assaad, SciPost Phys. 3 (2017), 013.

Lattice gauge theories

Confinement transition of $\mathbb{Z}_2$ gauge theories coupled to massless fermions: Emergent quantum chromodynamics and $SO(5)$ symmetry

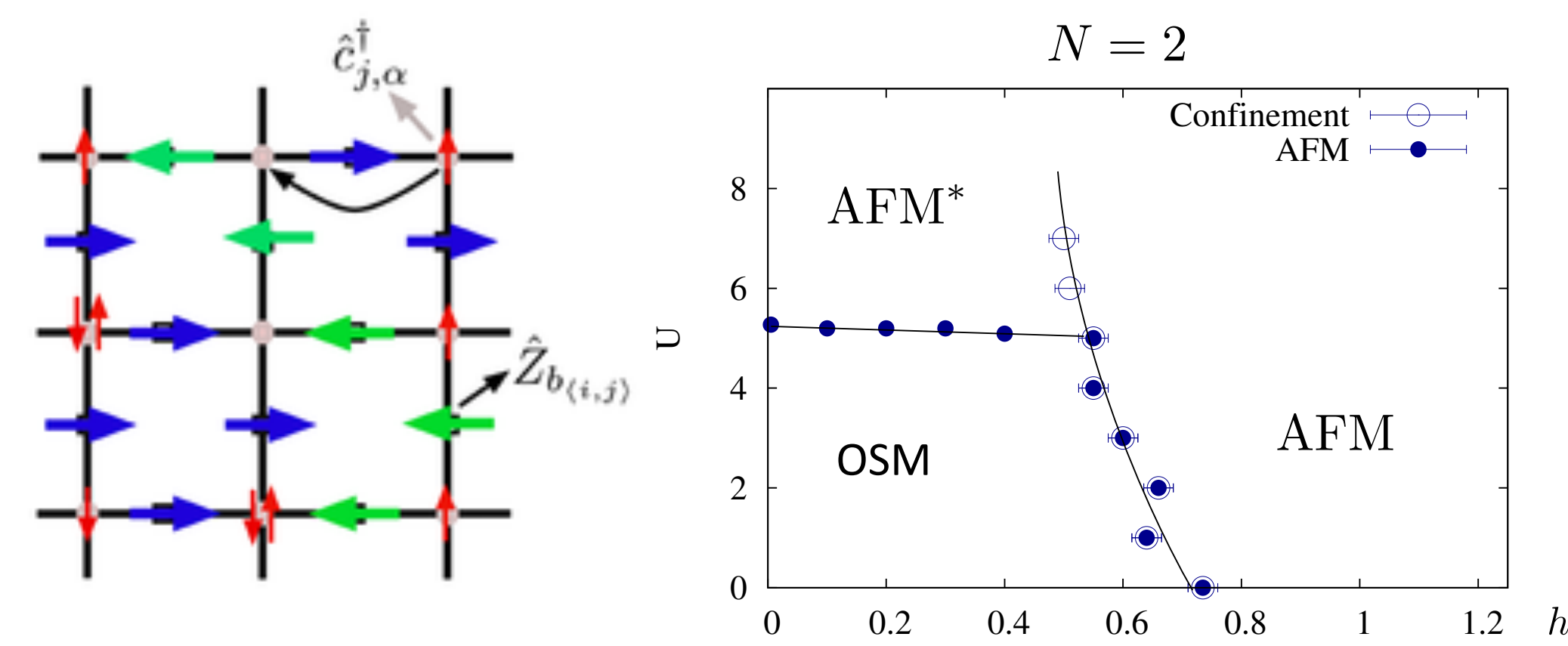
Snir Gazi^{a,1}, Fakhre F. Assaad^b, Subir Sachdev^{c,d,1}, Ashvin Vishwanath^c, and Chong Wang^c

PHYSICAL REVIEW X **6**, 041049 (2016)

Simple Fermionic Model of Deconfined Phases and Phase Transitions

F. F. Assaad¹ and Tarun Grover^{2,3}

$$\hat{H} = \sum_{\langle ij \rangle} \hat{Z}_{\langle ij \rangle} \left(\sum_{\alpha=1}^N \hat{c}_{i,\alpha}^\dagger \hat{c}_{j,\alpha} + \text{H.c.} \right) - N h \sum_{\langle ij \rangle} \hat{X}_{\langle ij \rangle}.$$



$$\hat{Q}_i = (-1)^{\sum_{\sigma} \hat{c}_{i,\sigma}^\dagger \hat{c}_{i,\sigma}} \hat{X}_{i,i+\hat{a}_x} \hat{X}_{i,i+\hat{a}_y} \hat{X}_{i,i+\hat{a}_z} \hat{X}_{i,i+\hat{a}_y} \hat{X}_{i,i+\hat{a}_x}, \quad [\hat{Q}_i, \hat{H}] = 0$$

The c-orthogonal fermion carries
global U(1) charge and local $\mathbb{Z}_2$ charge

	Gauge Field	Particle-hole
OSM	De-confined	No binding
AFM	Confined	Condensed
AFM*	De-confined	Condensed

See also.
M. Hohenadler and FFA (arXiv:1906.11937, PRL 18), Orthogonal metal in Falicov-Kimball model.
S. Gazit, FFA, and S. Sachdev (arXiv:1906.11250), OSM → FL* transitions

Intertwined orders in Dirac fermions

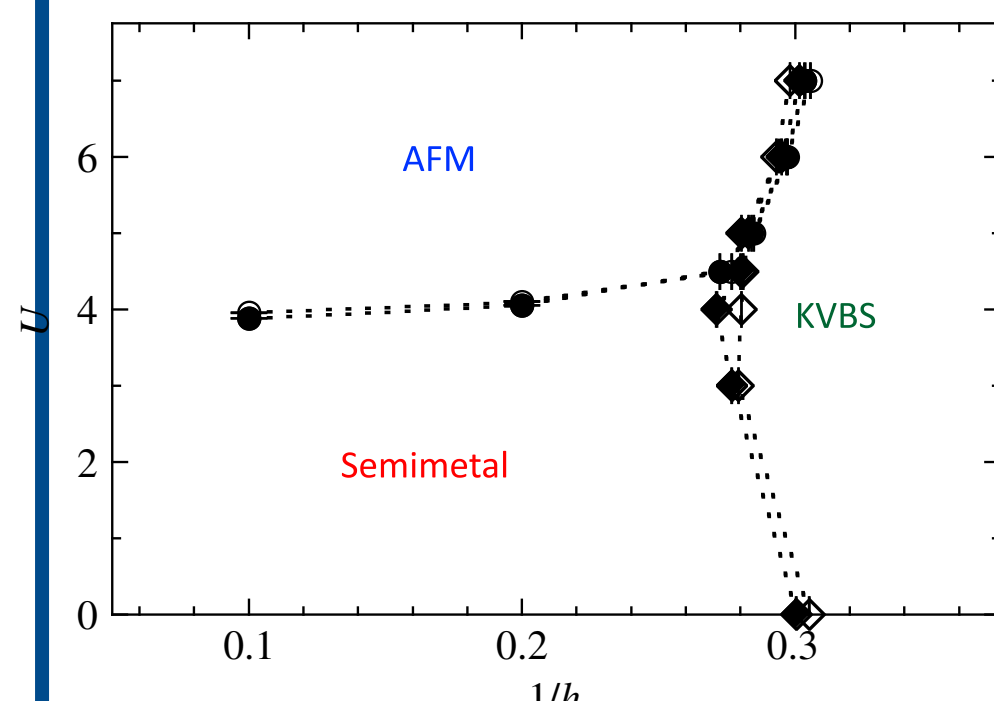
PRL **119**, 197203 (2017)

PHYSICAL REVIEW LETTERS

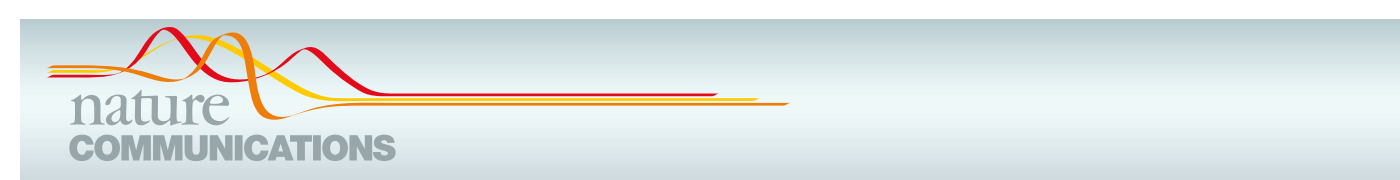
week ending
10 NOVEMBER 2017

Dirac Fermions with Competing Orders: Non-Landau Transition with Emergent Symmetry

Toshihiro Sato,¹ Martin Hohenadler,¹ and Fakhre F. Assaad¹



AFM (QSH) and VBS (SSC) from
quintuplet of anticommuting
Mass terms.



ARTICLE

<https://doi.org/10.1038/nrn4667-019-10372-0>

OPEN

Superconductivity from the condensation of
topological defects in a quantum spin-Hall insulator

Yuhai Liu¹, Zhenju Wang², Toshihiro Sato², Martin Hohenadler², Chong Wang³, Wenan Guo^{1,4} &
Fakhre F. Assaad²

$$\hat{H} = -t \sum_{\langle ij \rangle} (\hat{c}_i^\dagger \hat{c}_j + \hat{c}_j^\dagger \hat{c}_i) - \lambda \sum_{\langle \langle ij \rangle \rangle \in \mathcal{O}} \left(\sum_{\sigma} \nu_{i,j} \hat{c}_i^\dagger \sigma \hat{c}_j - \hat{c}_j^\dagger \sigma \hat{c}_i \right)^2$$

